Zishi technical documentation

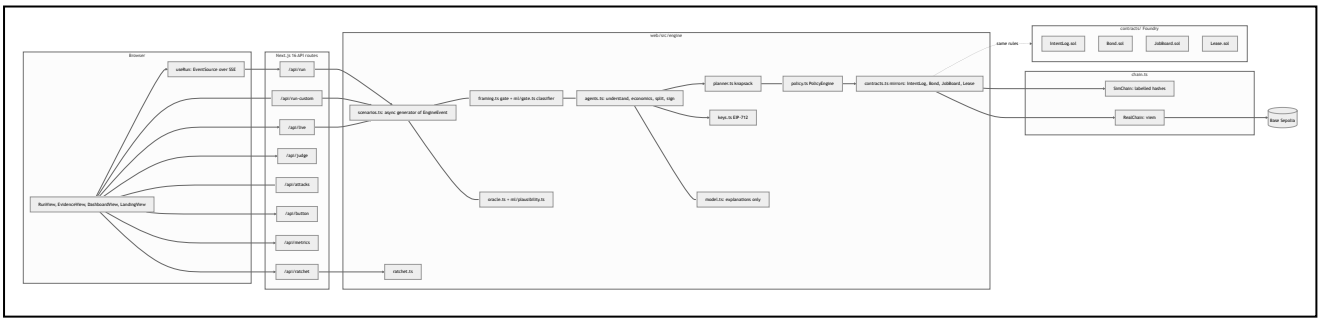
Zishi (自食其力) is a wallet for a machine. An agricultural spraying drone on hire purchase earns per mu sprayed, pays its lender from every fee, keeps a bond that is slashed only on cryptographic proof of a rule break, and owns itself when the lease reaches zero. The model explains; code decides. Everything below is grounded in `web/src/engine` and `contracts/`.

Companion documents: `docs/Zishi_Technical_Document.docx` (this material on the official submission template, with appendices), `docs/JUDGES_GUIDE.md` (ten minutes in order), `SELF_ASSESSMENT.md` (security risks and what is built), `docs/EXECUTION_EVIDENCE.md` (a real run checked against the chain), `docs/RED_TEAM.md` (the attack set, run at generation time), `docs/EVALUATION.md` (held out results, raw values in `docs/results/`), `DESIGN.md` (product design and economics), `contracts/README.md` (contract detail), `web/README.md` (how to run the app).

1. Problem

Who	Over 200,000 agricultural drone pilots in China. A drone costs about 55,000 yuan. Spraying pays 3 to 15 yuan per mu
Problem	A pilot with nothing to pledge cannot get a loan. The lender has no way to see the work or to collect from it
Zishi	The drone has its own wallet. Fees arrive per confirmed job, the instalment leaves by rule, the title moves at zero. A bond covers rule breaks so a counterparty can read before a job that the worst case is covered
Agent risk	The agent reads untrusted messages and signs intents. Every intent goes through a deterministic policy engine, and a broken rule in a signed intent is a proof that slashes the bond. The pilot never pays for the AI's mistakes: agent rules hit the vendor bond

2. Architecture



Part	File	Job
Scenario	`scenarios.ts`	One async generator per scenario (`fleet`, `neighbour`, `attack`). Takes { `fast, lang, real` }. Emits `EngineEvent` values in order. The UI draws what it receives and computes nothing
Gate	`framing.ts`, `ml/gate.ts`	Escapes and wraps every inbound message, applies pattern rules, adds a classifier risk score
Agent	`agents.ts`, `model.ts`	Reads the message into typed fields, computes economics, signs intents. The model writes one sentence explanations and reads messages into a schema; it never names a payee or an amount
Planner	`planner.ts`	Bounded knapsack over the day's candidate jobs

Part	File	Job
Policy engine	`policy.ts`	Evaluates every signed intent against rules that are data. Allow, deny or escalate
Mirrors	`contracts.ts`	TypeScript copies of the four contracts with the same rules and numbers, so the demo runs with no chain
Oracle	`oracle.ts`, `ml/plausibility.ts`	Checks a flight log by geometry and by a robust z anomaly model
Ratchet	`ratchet.ts`	Hill climbing over the acceptance strategy against a frozen week
Keys	`keys.ts`	One secp256k1 key per actor, generated at boot. EIP-712 domain `Zishi` v1, chain 84532
Chain	`chain.ts`	`SimChain` or `RealChain` behind one `Chain` interface
Contracts	`contracts/src`	Solidity 0.8.26, Foundry, 68 tests, deployed on Base Sepolia

3. Data model

All money is yuan with two decimals. Addresses are `0x` hex.

3.1 Events (`types.ts`, `EngineEvent`)

t	Fields	Meaning
`narration`	`text, at`	A line of story
`agent.message`	`from, to, text, untrusted, facts?, source?, grounded?`	A message between actors. `source` is `model` or `template`; `grounded` says every number came from the facts
`gate`	`from, action, categories, reason, framed, risk?`	The gate verdict on an inbound message: `pass`, `redact`, `hold`, `refuse`
`understood`	`jobId, fields, source, confidence?`	The message read into `{ task, window, fee, geofence, payeeName }` by the model or by rules
`plan`	`items, budgetMu, usedMu, charges, chargesUsed, totalMargin, usedMinutes, method`	The day plan from the knapsack
`solvency`	`jobId, worst, bond, min, floor, covers, earnIn`	Bond plus floor against a quarter of the fee
`economics`	`jobId, eco`	Fee, instalment, energy, rail, margin, reserve rule, clears
`telemetry`	`jobId, trackM, swathM, seconds, areaMu, claimedMu, ratio, ok, batteryPct, altVar, z, reasons`	A flight log and its anomaly score
`log`	`jobId, intentHash, parties`	The intent hash committed to the log and the full intent delivered to every party
`intent`	`intent`	A signed intent (below)
`policy`	`decision`	`{ intentId, verdict, lines[], signer }`

t	Fields	Meaning
`chain`	`tx`	A transaction, emitted up to three times under one hash: `pending`, `confirmed`, `finalised`
`proof`	`intentId, intentHash, rule, signer, signatureOk, slash, bounty, treasury, bond, shadow, submitter, submitterAddr`	A slash by proof
`machine`	`state`	The drone's full state (balances, bond, lease, position, status)
`job`	`job`	A job and its status
`attack`	`name, blocked, where, detail, id, standard, naive, isolated`	One attack case result
`button`	`waiting, what`	The physical button is needed or has been pressed
`done`	`summary`	End of run

3.2 Intent (`Intent`)

Field	Type	Note
`id`	string	Unique per intent
`actor`	Actor	`machine`, `operator`, `requester`, `lender`, `hunter`, `vendor`, `oracle` sign; others do not
`kind`	enum	`pay`, `accept_job`, `start_task`, `claim_done`, `pay_instalment`, `set_cap`, `top_up`
`payee`	address	Payments only. Always from the allowlist, never from message text
`amount`	number	Yuan, two decimals
`jobId`	string	Maps one to one to a `JobBoard` job id
`geofence`	string	The field the task runs in
`confirmed`	bool	On `claim_done`, what the agent asserts. The contract checks the board, not this field
`nonce`	number	Replay protection
`signer`	address	Recovered from the signature
`signature`	hex	secp256k1 over the EIP-712 digest. Same digest as `Bond.digestOf` on chain
`provenance`	`{ amount, payee }`	Where each came from: `tool`, `allowlist`, `message`, `none`. A `message` source denies

3.3 Job (`Job`)

Field	Type	Note
`id`, `side`, `title`	string	`side` is `neighbour`, `fleet` or `attack`
`fee`, `mu`, `ratePerMu`	number	Fee in yuan, area in mu
`requester`, `geofence`, `window`	string	Who posted it, the field, the time window

Field	Type	Note
`status`	enum	`posted`, `locked`, `running`, `confirmed`, `released`, `held`, `declined`
`shadow`	bool	Shadow jobs move no lease money; hunters report only on these
`split`	`{ owner, instalment, rail, bond }`	Filled on release
`hold`, `holdUntil`	number	The flaggable hold (13.33 percent of the fee, 24 hours)
`from`, `to`	xyz	Positions for the map

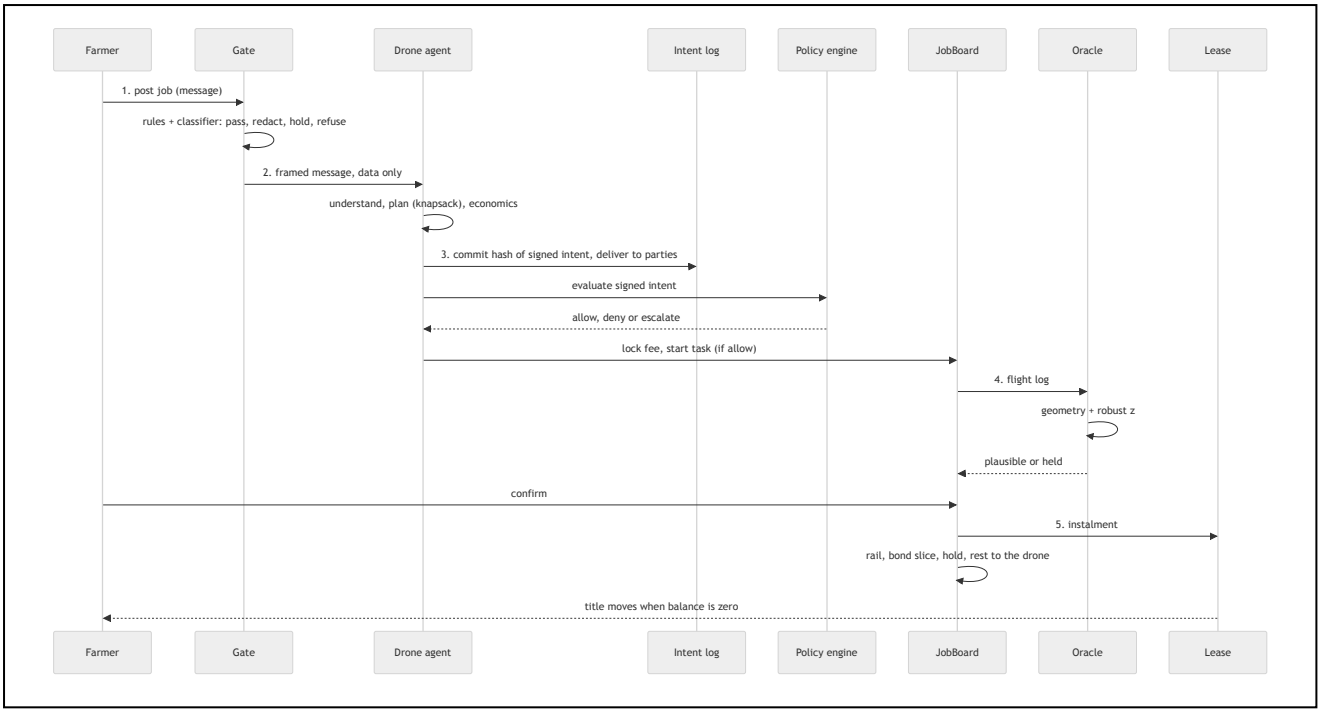
3.4 Bonds and lease (`MachineState`, `Bond.sol`, `Lease.sol`)

Field	Where	Note
`bond`	pilot bond	Earned from a slice of each fee. Slashed only by `FalseConfirmation` and `OutsideGeofence`
`bondMin`	reward threshold	2,000 yuan. Reaching it earns a better rate and a board listing; never a start condition
`startFloor`	start check	200 yuan, added to the pilot bond. A zero bond can start jobs up to 800 yuan in fee
`vendorBond`	vendor bond	5,000 yuan posted by the vendor. Slashed only by `PayeeNotAllowlisted` and `OverCap`, penalty 200, daily cap 1,000
`leaseBalance`, `leaseTotal`	lease	Instalment per released fee is `min(fee * rate, balance)`, rate in the signed band 20 to 45 percent
`muDone`, `muTotal`	lease	13,750 mu across the demo lease
`owned`	title	True once the lease balance is zero

Demo numbers live in `DEMO` in `policy.ts`.

4. The loop

Every job runs the same five phases. The live demo shows five moments, one per failure or success path.



Moment	Phase	What the demo shows
1	Plan	A job declined: margin per mu below the reserve rule
2	Confirm and split	A paid flight: farmer confirms, fee splits, instalment reaches the lender
3	Oracle	A faked flight log, caught by the anomaly model, fee held
4	Proof	A compromised intent slashed from the vendor bond by proof; the harmed party is paid
5	Title	Lease at zero, title moves to the pilot

5. Policy engine

`PolicyEngine.evaluate(intent, ctx)` in `policy.ts`. Rules are data in `DEFAULT_POLICY`. Each line carries a `tier`. A failed `operator only` line on a valid machine intent escalates to the button; any other failure denies.

Rule	Applies to	Check	Tier
`signature`	all	Recovered signer equals the actor's key	L2
`replay`	all	Signature hash never seen before	L2
`amount`	money kinds	Finite, positive, two decimals	L2
`provenance`	pay, pay_instalment	Neither amount nor payee came from message text	L2
`allowlist`	pay, pay_instalment	Payee is on the allowlist	L2
`lookalike`	pay, pay_instalment	Payee within 3 characters of an allowlisted address is named and denied	L2
`instalment payee`	pay_instalment	Payee is the lender	L2
`per transaction cap`	pay, pay_instalment	Amount at most 1,000 yuan	L2
`daily cap`	pay, pay_instalment	Spent today plus amount at most 5,000 yuan	L2
`rate limit`	pay, pay_instalment	Fewer than 12 payments in the last hour	L2

Rule	Applies to	Check	Tier
`deviation`	pay, pay_instalment	Within 25 percent of the reference amount	L2
`rate band`	pay_instalment	Instalment share of the fee between 20 and 45 percent	L2
`geofence`	accept_job, start_task	Field is one of the known geofences	L2
`contact`	accept_job	Requester is a known contact	L2
`listing band`	accept_job	Fee inside the listing's band	L2
`economics`	accept_job	Margin clears the reserve rule	L2
`job`	start_task	Job exists and is `locked`	L2
`job geofence`	start_task	Intent geofence equals the job's geofence	L2
`solvency`	start_task	Bond plus floor covers a quarter of the fee	L2
`confirmation`	claim_done	The agent asserts confirmed	L2
`operator only`	set_cap, top_up	Signed by the operator key	L3

Tiers: L0 read (anyone, the model), L1 draft (the model writes text, never a payee or amount), L2 engine allow (machine key through the rules above), L3 operator (the physical button, one action at a time). The model never sees this code. Tiers are checked on the signed intent, so a jailbroken model cannot move one.

6. Gate and learned components

6.1 Gate (`framing.ts`)

Every inbound message is escaped and wrapped in a `relay-message` block with `relation` (owner, counterparty, unknown) and `authority` attributes. Only the owner's messages carry `authority="instruction"`. `judge(body, relation)` runs in this order:

Order	Condition	Action
1	Body over 64 KiB	refuse
2	Asks for a key	refuse
3	Tries to change the agent's instructions	refuse
4	Names a payee, cap or location in the text	redact
5	Asks for money or to skip a rule	hold
6	Pressure words only	pass, noted
7	Rules pass but classifier `p > 0.8`	hold, "classifier flagged"
8	Otherwise	pass, as data

The classifier can only turn a pass into a hold. The rules keep the last word on refuse.

6.2 Learned and optimising components

Each one has a fixed seed and a held out set the agent cannot edit. `pnpm test` prints the numbers, `pnpm bench` prints the before and after table below, `/api/metrics` serves them, and the Evidence page shows them under Learned components and Monitors.

Component	File	Borrowed from	Method	Data and seed	Held out result
Message classifier	`ml/gate.ts`, `ml/corpus.ts`	Text classification	Logistic regression on hashed character 1, 2 and 3-grams, 2^14 buckets, trained at module load, adds a risk score the rule gate can only turn into a hold	401 messages in English and Chinese, 100 held out, seed 20251020	accuracy 0.95, precision 0.93, recall 0.96, F1 0.94, false alarm 0.054
GPS fusion	`fusion/kalman.ts`	Aerospace, the Kalman filter	Constant velocity Kalman filter with a Rauch Tung Striebel smoother over 1 Hz GPS samples; measurement noise estimated per flight; innovation gating rejects outliers; an FNV hash of the innovation sequence detects a replayed track	the same 2,000 synthetic flights, each with a noisy track	at matched full detection the raw track length false alarms on 13.8 percent of honest flights, the smoothed length on 0.0 percent
Anomaly threshold	`fusion/conformal.ts`	Statistics, split conformal prediction	The flag threshold is the calibration set quantile for a 2 percent target false alarm rate, so the guarantee is stated and checked rather than hand set	634 honest calibration flights, 500 held out, seed 7331	threshold z 2.60, held out false alarm 3.0 percent, coverage 97 percent, detection 100 percent on all five tamper classes
Telemetry anomaly model	`ml/plausibility.ts`	Robust statistics	Robust z (median and MAD) over area ratio, speed, battery per mu, altitude variance and GPS innovation, fitted on 750 flights	2,000 flights, 15 percent tampered across inflated area, replayed track, impossible speed, low battery and teleport jump	72 of 72 tampered flights caught

Component	File	Borrowed from	Method	Data and seed	Held out result
Pilot credit score	`credit/score.ts`	Actuarial science, Buhlmann credibility	A Beta posterior per component (confirmations, instalments on time, clean record) weighted n to n plus k against the fleet prior; the score moves the instalment share inside the policy band and is emitted as a `credit` event	fleet prior strength 8	0.912 to 0.936 after three confirmed jobs, 0.882 after a held log; tier B at 2,815 bps
Day planner	`planner.ts`, `economics/pricing.ts`	Operations research and mechanism design	Bounded knapsack over mu per charge, charges and minutes; overlapping jobs settle by a second price comparison so an inflated stated fee cannot buy priority	the day's candidate jobs	exact; a maize job that inflates its bid to 5 wins at 4.2 and loses 0.7 in utility
Lease MDP	`economics/mdp.ts`	Operations research, dynamic programming	Finite horizon Markov decision process over lease balance, bond, weather and offer class, 46,800 states, value iteration over 26 weeks, compared against the greedy rule with common random numbers	200 seeded seasons, seed 2026	7,540 versus 7,471 yuan mean profit, same weeks to title
Strategy ratchet	`ratchet.ts`	The autoresearch loop	Hill climbing over four acceptance knobs against a frozen simulated week, keeping a change only if profit after instalments rises and the reserve rule holds	seed 4242, walk seed 99	60 steps, 5 kept, 246 to 926 yuan

Component	File	Borrowed from	Method	Data and seed	Held out result
Bond stress test	`economics/stress.ts`	Bank risk, Monte Carlo	2,000 seeded seasons with tamper attempts, slashes and skipped instalments; reports default probability, bond drawdowns and lender loss given default	seed in `STRESS_DEFAULTS`	default 9 percent; p95 loss 25,593 yuan without bonds, 19,739 with, bonds absorb 23 percent
Drift monitor	`drift/cusum.ts`, `drift/monitor.ts`	Manufacturing, statistical process control	One sided CUSUM on the classifier's risk scores and on the policy deny rate, alarm resets, 120 point history for the sparklines	synthetic shift at observation 500, seed 20251020	alarm 6 observations after the shift, 0 false alarms in 600 benign scores
Replay log	`replay/log.ts`	Financial ledgers, event sourcing	Every event of a run appended to a hash chain with volatile fields dropped from the canonical form; `GET /api/replay` lists and verifies	two seeded fast runs	identical chain head over 293 events, a tampered entry caught at its position

What they are not: not trained on live traffic or real drone logs. The classifier and the anomaly model only raise holds. The planner only orders work. The ratchet and the MDP only move acceptance knobs inside the policy band. The credit score only recommends a rate inside the band. Nothing here signs, pays or changes a cap.

6.4 Measured improvement

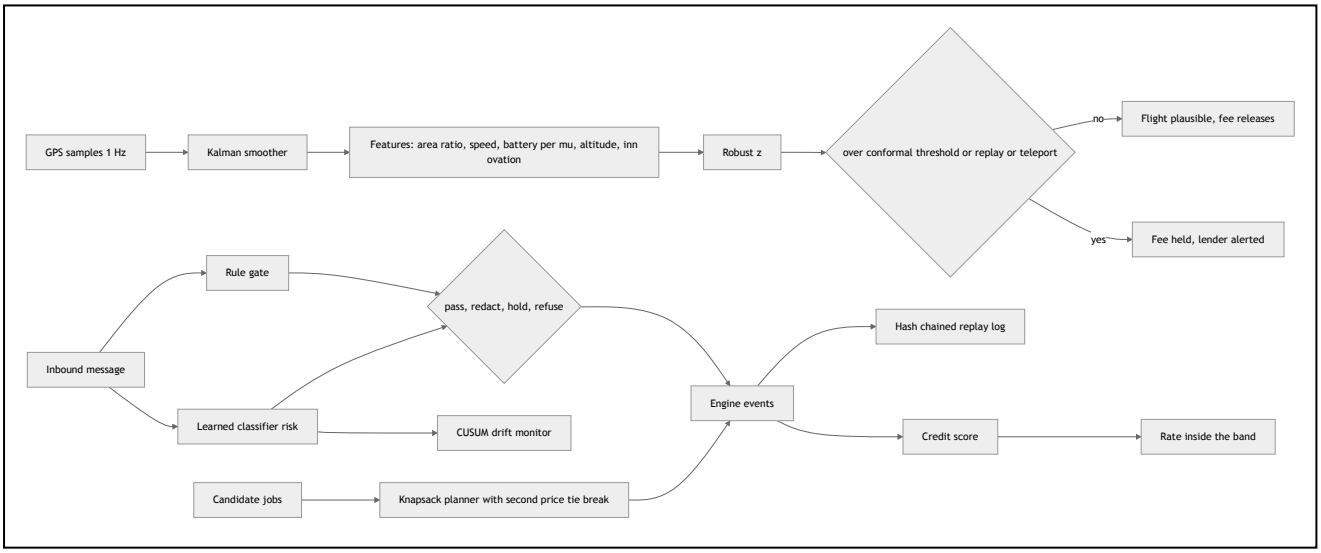
`pnpm charts` (`web/scripts/charts.ts`) redraws nine SVG charts in `docs/charts` from the same seeds: the detection curve, the robust z histogram, conformal calibration, lender loss against skip rate, MDP against greedy across balances, the ratchet walk, three credit trajectories, the CUSUM trace and the classifier across thresholds. The README shows them.

`pnpm bench` recomputes every row from the frozen seeds. Nothing in this table is pasted.

Component	Before	After	Before result	After result
Message gate	rules only (regex)	rules plus learned classifier	52 of 52 scripted attacks, no score on paraphrases	held out 87 messages, no attack text in training: accuracy 93.1%, recall 81.3%, false alarm 0 of 55
Telemetry check	raw GPS length	Kalman smoothed	track features only: false alarm 13.8% at full detection	0.0% at full detection; the oracle's own rate with all features is 3.0% (13 of 428)

Component	Before	After	Before result	After result
Anomaly threshold	fixed	z	4	conformal
Lease strategy	greedy rule	MDP policy (26 weeks, value iteration)	7,471 yuan mean, title 84.0%	7,540 yuan mean, title 84.0%
Acceptance knobs	baseline	ratchet, 5 of 60 steps kept	246 yuan	926 yuan
Lender loss given default, p95	no bonds	pilot and vendor bonds	25,593 yuan	19,739 yuan, bonds absorb 23.0% (default 9.0%)
Pilot credit	flat rate band	Buhlmann credibility	no change with history	p 0.936 after 3 confirmed, 0.882 after a held log
Drift detection	none	CUSUM on classifier risk	never	alarm 6 observations after the shift, 0 false alarms in 600
Run integrity	no log	hash chained event log	not verifiable	two seeded runs, head 0x68c3abcf, equal true

How to read it. The Kalman row holds detection at 100 percent on the track features and compares false alarms, which is the cost a pilot feels. The conformal row trades a hand set threshold that happened to be safe on this data for one with a stated target and a measured held out rate. The MDP and ratchet rows are profit on a frozen simulated season and week, not on real farms. The stress row is the lender's view: the bonds take about a quarter of the tail loss, and the title takes the rest.



6.3 Model (`model.ts`)

With `AI_GATEWAY_API_KEY` or `ANTHROPIC_API_KEY` the model writes explanations and reads messages into a Zod schema with a confidence. Without a key, templates write and rules read. In both cases:

- the model receives a JSON object of numbers the tools computed;
- the sentence that comes back is checked and rejected if it contains a number not in the facts;
- a timeout, error or failed check returns the template;
- `model.ts` has no key, no chain client and no signing function.

7. Contracts

Solidity 0.8.26, Foundry, 68 tests in `contracts/test` (15 of them the contract layer attack cases ported from the engine's attack set, 9 for Lease), three of them fuzz tests over the slash path (amount inside the cap never slashes, any amount over the cap slashes exactly the penalty, any payee off the allowlist slashes). Amounts in the tests use 1 ether as 1 yuan. Deploy order: `IntentLog`, `Bond(treasury, log, floor)`, `Lease`, `JobBoard(bond, lease, rail)`, then `bond.setBoard(board)`.

Deployed on Base Sepolia (chain 84532) from the demo wallet, source verified on Sourcify with an exact match: IntentLog and Lease in block 48,335,911, Bond and JobBoard redeployed in block 48,339,163 after the collusion fix. Addresses and deploy transactions are in `contracts/deployments/base-sepolia.json`; the script is `contracts/script/Deploy.s.sol`.

Contract	Address
IntentLog	`0x093eD0DD8E4bE4b0205a6e39d0c5988f5Cd8ad51`
Bond	`0x6161f6e08c8695849e7A8aF74c39a5A1056d5435`
Lease	`0x0D21c8f6C1e6e6EA2ae5fF073F51fC4d57AC5ed2`
JobBoard	`0xAbE83C0dD54EAd67DB63C8f9e854f00756f29952`

7.1 IntentLog

Function	Who	Does
`commit(jobId, intentHash)`	agent key	Records hash, job and sender once
`committedAt(hash)`, `jobOf(hash)`, `agentOf(hash)`	view	Read an entry

7.2 Bond (implements IBonded)

Function	Who	Does
`postVendorBond(penalty, dailyCap)` payable	vendor	Posts or tops up the vendor bond and its terms
`register(agent, operator, vendor, minimum, perTxCap, slashAmount, allow[], agentSig)`	operator	Registers an agent; the vendor must hold a bond
`fill(agent)` payable	JobBoard	Adds a fee slice to the pilot bond
`topUp(agent)` payable	operator	Tops up the pilot bond
`bondOf(agent)`, `worstCase(agent)`, `covers(agent, exposure)`	view	`covers` is `balance + startFloor >= exposure`
`vendorBondOf`, `vendorOf`, `operatorOf`, `vendorTerms`, `isAllowlisted`	view	Reads for the board and the UI
`hashIntent`, `digestOf`, `DOMAIN_SEPARATOR`	view	EIP-712, same digest as `keys.ts`
`slashByProof(intent, signature, rule)`	a party of the job, or a hunter on a shadow job	Checks the proof and slashes the tagged bond
`setBoard(board)`, `setHunter(hunter, on)`	deployer	One time wiring, hunter registry

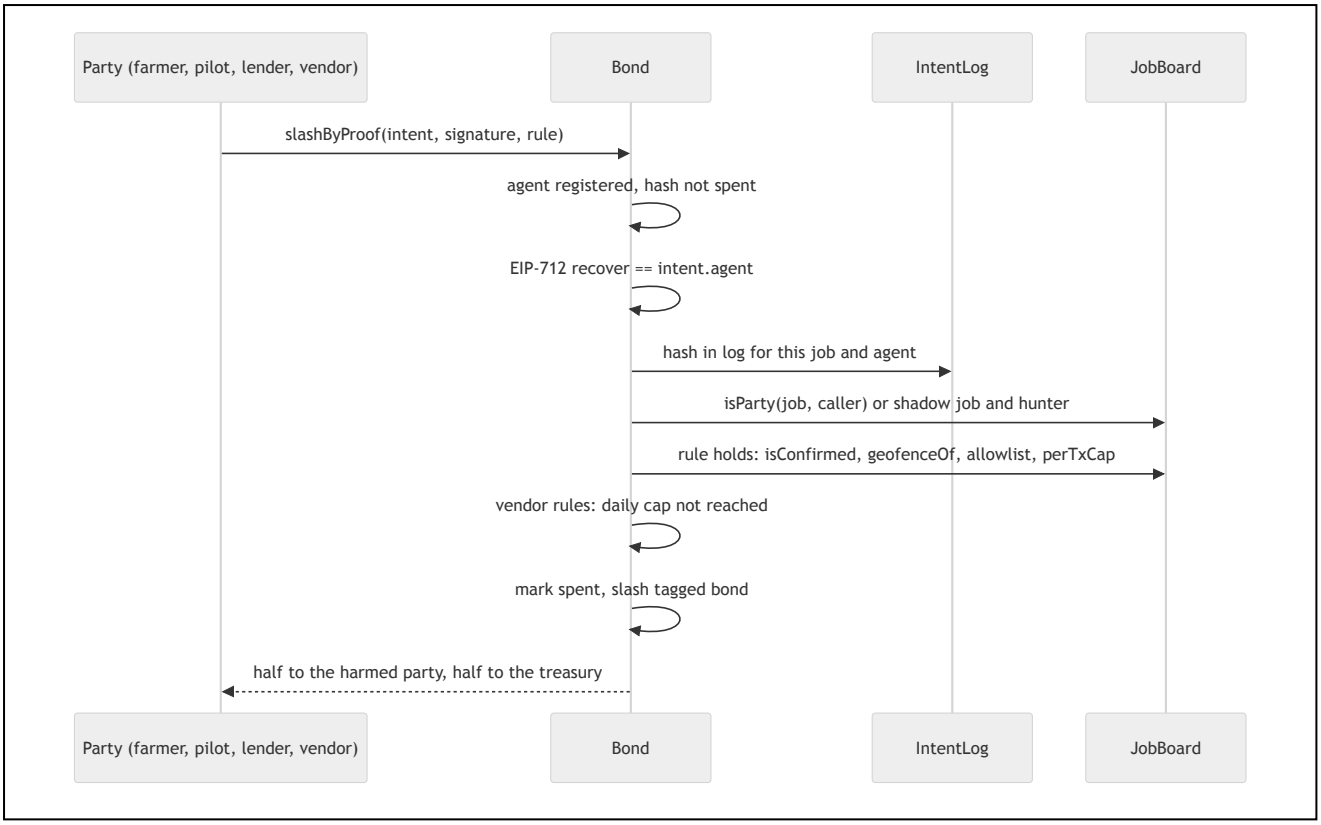
7.3 JobBoard

Function	Who	Does
`post(agent, verifier, leaseId, geofence, windowStart, windowEnd, shadow)` payable	requester	Locks the fee, status `Posted`
`start(id)`	agent	Needs `bond.covers(agent, fee / 4)`; status `Running`
`confirm(id)`	verifier	Splits the fee: rail 2 percent, hold 13.33 percent for 24 hours, instalment to `Lease.pay`, bond slice `min(gap, rest / 2)` to `Bond.fill`, rest to the agent
`flag(id)`	requester	Inside the hold window: refunds the hold, the rest goes to a human arbiter
`releaseHold(id)`	anyone	After the window, pays the hold to the agent
`cancel(id)`	requester	A never started job after its window; fee refunded
`isConfirmed`, `geofenceOf`, `isParty`, `isShadow`, `requesterOf`	view	Read by `Bond.slashByProof`

7.4 Lease

Function	Who	Does
`open(operator, agent, total, rateBps, bandLoBps, bandHiBps, monthlyMinimum)`	lender	Opens a lease
`pay(id)` payable	JobBoard	Instalment `min(fee * rate, balance)`; emits `TitleTransferred` at zero
`instalmentFor(id, fee)`, `isFrozen(id)`, `lenderOf(id)`	view	Reads
`freeze(id, reason)`, `unfreeze(id)`	lender	Freezes instalment flow
`setRate(id, rateBps)`	operator	Inside the signed band

7.5 Proof of violation



Rule	Bond	Violation	Read from
`PayeeNotAllowlisted` (0)	vendor	pay or pay_instalment to a payee not on the allowlist	intent, `Bond`
`OverCap` (1)	vendor	amount over `perTxCap`	intent, `Bond`
`FalseConfirmation` (2)	pilot	claim_done asserting confirmed while `JobBoard.isConfirmed` is false	intent, `JobBoard`
`OutsideGeofence` (3)	pilot	start_task with a geofence different from `JobBoard.geofenceOf`	intent, `JobBoard`

Revert order: `unknown agent`, `proof already used`, `not the agent's signature`, `not in log`, `not a party`, `no violation`, `daily cap reached`. Nothing comes from the reporter's word. On a live job the submitter earns no bounty, so compromising a live wallet pays nothing; hunters earn half of the slash on shadow jobs only. The engine mirror `BondContract.slashByProof()` applies the same checks and numbers.

8. Chain modes

Mode	When	What a transaction is
Simulated	Default	`SimChain`: keccak hash of the event, `simulated: true` on every tx, labelled in the UI
Real	`SURETY_PRIVATE_KEY` set, wallet funded, and the request carries `x-zishi-demo: <DEMO_TOKEN>`	`RealChain`: every signed intent is `IntentLog.commit(jobId, intentHash)` on the deployed contract (69,288 gas); every money event is a self transfer on Base Sepolia with the event JSON as calldata, label `anchor: ...` (28,880 gas). Explorer link on both, confirmed at one receipt, finalised at two more blocks

The demo token arrives once as `?demo=TOKEN` on the app URL, is kept in the browser, and is sent as a header on every run (`components/demoToken.ts`). Without it the funded key is never used, so a public link

cannot drain it. `/api/metrics` reports `chain: "real"` only when the wallet holds testnet ETH (`realChainFunded`, checked at most once a minute). The four contracts are deployed on Base Sepolia (section 7). In real mode every signed intent is committed to the deployed IntentLog by a contract call (`RealChain.commit`, emitted as a `chain` event of kind `log` with the explorer link); a hash the log already holds falls back to a calldata anchor and says so. Money events are anchored as calldata, and the Bond, JobBoard and Lease rules run in the mirrors; their calls are the next step.

One wallet signs for the whole demo, and a run sends transactions a few hundred milliseconds apart, so `RealChain` sends one at a time through an in-memory queue, fetches the pending nonce for each, and retries up to three times on a nonce clash from another instance on the same key (`nonce too low`, `replacement transaction underpriced`). A commit is simulated first (`simulateContract`), so a hash the log already holds costs no gas and is anchored with the reason in its label. A full real run of the fleet scenario is 15 commits and about 30 anchors, 0.000013 ETH at 0.006 gwei.

9. Bring your own data

The scripted scenarios are one way in. The demo page (`/app/fleet`) runs the same engine on other data in two more modes, Live feed and Your data. Nothing else changes: gate, planner, policy engine, intent log, mirrors and chain adapter are the ones above.

Mode	What happens
Live feed	`GET /api/live` streams a new job every few seconds from a seeded generator (`engine/generate.ts`): rice, maize and passion fruit plots at market prices, and every so often an injection attempt at the wallet. Same seed, same feed, so a session can be replayed. Runs until you stop
Your data	Drop a JSON or CSV file, paste it, or pick a sample. `POST /api/run-custom` validates it, builds the run and streams the same `EngineEvent` values as `/api/run`

9.1 Schema (`engine/schema.ts`)

Validation is Zod. A bad row fails the whole request with the path and the reason (`jobs.2.mu: must be more than 0`). Amounts are yuan, areas mu, distances metres.

Object	Field	Required	Rule
job	`id`	yes	1 to 40 chars, letters, digits, dot, dash, underscore; unique
job	`title`	yes	1 to 120 chars
job	`mu`	yes	0 to 10,000
job	`pricePerMu`	yes	0 to 1,000,000
job	`field`	yes	The geofence name, 1 to 40 chars
job	`window`	no	Free text, default `today`. `08:00–12:00` is read as 240 minutes for the planner
job	`requester`	yes	A farmer's display name, 1 to 60 chars
job	`distanceM`	no	0 to 50,000, for the planner
flight	`jobId`	yes	Must match a job id
flight	`trackM`, `seconds`, `swathM`, `claimedMu`	yes	Positive; swath at most 50 m, time at most a day

Object	Field	Required	Rule
flight	`batteryPct`, `altVar`	no	Default to what the track predicts. Without a flight the engine synthesises an honest log for the job's area
message	`from`	yes	Sender's display name
message	`text`	yes	Up to 64 KiB. With a `jobId` it replaces the generated offer for that job; without one it is an unsolicited message through the gate
policy	`perTxCap`, `dailyCap`, `maxPerHour`, `deviationPct`, `rateBand`, `allowlist`, `contacts`, `geofences`	no	Overrides for `DEFAULT_POLICY`. By default every requester and field in the input is trusted; set `contacts` or `geofences` to see a decline
lease	`total`, `rate`, `floor`, `balance`	`total`	Drone price, instalment share (0.01 to 0.9, default 0.4), reserve rule (default 30), amount still owed (default `total`)
run	`jobs`, `flights`, `messages`, `policy`, `lease`, `lang`, `seed`	`jobs`	1 to 50 jobs, up to 50 flights and 50 messages, `lang` `en` or `zh`, `seed` for positions and generated logs

9.2 Files (`lib/ingest.ts`, `lib/csv.ts`)

Input	Shape
JSON	The run object above, or a bare array of jobs
Jobs CSV	Header row then one job per line. Columns: `job id` (or `id`), `title`, `mu`, `price per mu`, `field`, `window`, `requester`, `distance m`. Header names match loosely: `Price per mu`, `price_per_mu` and `pricePerMu` are the same column. Quoted fields, doubled quotes, CRLF and a BOM are handled
Flights CSV	Optional second file: `job id`, `track m`, `seconds`, `swath m`, `claimed mu`, `battery pct`, `alt var`

Samples are listed by `GET /api/samples` and appear in the page's dropdown.

9.3 Endpoints and limits

Endpoint	Request	Response
`POST /api/run-custom?lang=en`	`application/json` with the run object, or `multipart/form-data` with `file` (JSON or jobs CSV) and optional `flights` (CSV)	SSE stream of `EngineEvent`; on a bad input `400 { issues: [{ path, reason }] }`
`GET /api/live?scenario=fleet&lang=en`	none	SSE stream that keeps producing jobs until the client closes it
`GET /api/samples`	none	`{ samples: [{ name, path }] }`, each path a static file the page can fetch

Limit	Value
Jobs, flights, messages	50 each per run
Message text	64 KiB, the gate's limit
Rate	Best effort, per server instance: 10 requests per minute per IP, one open stream per IP
Chain	Simulated unless the request carries the demo token

Limit	Value
Storage	None. Rows live for the request. Message text reaches a model only when a model key is set

10. API reference

All routes are dynamic. Limits: 10 requests per minute per IP and one concurrent run per IP (`ratelimit.ts`), `429 { error: "too many requests" }` when exceeded. The counters live in the memory of one server instance, so on a serverless host they are best effort; the funded key is guarded by the demo token, not by this limiter. `/api/metrics` is served from the CDN for a minute.

Method, path	Params	Response
`GET /api/run`	query `scenario` (`fleet`, `neighbour`, `attack`; default `fleet`), `fast=1`, `lang` (`en`, `zh`); header `x-zishi-demo` for real chain	`text/event-stream`, each line `data: <EngineEvent>`. Up to 300 s
`POST /api/run-custom`	query `lang`; body JSON run object or multipart `file` plus optional `flights` (section 9)	SSE as above, or `400 { issues }`
`GET /api/live`	query `scenario`, `lang`	SSE, a generated job every few seconds until closed
`GET /api/samples`	none	`{ samples: [{ name, path }] }`
`POST /api/judge`	JSON `{ text }`, text up to 70,000 chars	`{ gate: { action, categories, reason, risk }, framed, understood: { action, payeeName, amount, source, confidence, droppedAddress }, shadow: { intent, decision } | null, naive: { paid, payee, amount }, model, liveWalletMoved: false }`
`GET /api/attacks`	none	`{ results: AttackRow[], benign: { passed, total }, naive: { fell, total }, ranAt }`
`POST /api/button`	JSON `{ what: "top_up" | "set_cap", amount, scenario, lang? }`	`{ events: EngineEvent[] }`; `404` when no run exists for that scenario, `400` on an unknown action
`GET /api/metrics`	none	`{ gate, plausibility, ratchet: { baselineProfit, bestProfit, steps, kept, best, baseline, evalSeed, walkSeed }, planner: { method }, chain: "real" | "sim", model: "real" | "scripted", computedAt, note }`
`GET /api/ratchet`	query `steps` (default 60)	`{ steps[], best, baseline, baselineProfit, bestProfit, evalSeed, walkSeed, days, offers, ranAt, note }`, cached per `steps`

`AttackRow`: `{ id, name, nameZh, layer, standard, isolated?, blocked, detail, detailZh?, naive }`. `standard` is one of the five rubric items: injection defense, permission hierarchy, hallucination prevention, intent understanding, task planning.

11. Evidence and verify

Script	Command	Does
`scripts/evidence.ts`	`pnpm evidence`	Runs `fleet` and `neighbour` fast, writes `evidence/run-<stamp>.json` (one row per chain claim: kind, amount, from, to, hash, block, simulated, explorer), `run-<stamp>.state.json` (initial and final machine state, every proof) and `transcripts/run-<stamp>.jsonl` (every model call). `JIXIN_REAL=1` with a funded key anchors on Base Sepolia
`scripts/verify.ts`	`pnpm verify`	Reads the latest run, recomputes the drone's balance from the chain feed and compares it to the final state; for every anchored hash fetches the transaction from Base Sepolia, decodes the calldata and checks kind, amount, from and to. Exit 1 on any mismatch
`engine/selfcheck.ts`	`pnpm test`	Every attack blocked, every benign message passing, a normal instalment allowed, every slash rule has a contract case, removed rules absent, the model layer has no reach into keys or chain, learned metrics at or above their floors

The Evidence page in the app runs the attack set at request time and reads `/api/metrics`. Nothing on it is typed in.

12. Internationalisation

Dictionaries	`web/src/i18n/en.ts` and `zh.ts`, same keys, Simplified Chinese
UI	Every string through `useLang()` from `LangProvider.tsx`; `LangToggle.tsx` switches
Engine	`engine/i18n.ts`: `tr(lang, key, vars)` and `both(key, vars)`. Policy lines carry `detail` (English) and `detailZh`; attack rows carry `nameZh` and `detailZh`
Routes	`/api/run` and `/api/button` take `lang`; narration and agent messages come back in that language
Rule ids	Always English, stable in the JSON

Add a key to both files or the build fails the type check.

13. Deployment

Vercel, Next.js 16, Node 22, pnpm 10. Root directory `web`. The route `/api/run` sets `maxDuration = 300`.

Variable	Required	Note
`AI_GATEWAY_API_KEY`	no	Vercel AI Gateway key. Model writes explanations and reads messages. Without it: templates and rules
`ANTHROPIC_API_KEY`	no	Direct Anthropic key, same effect
`SURETY_PRIVATE_KEY`	no	`0x` plus 64 hex. Funded Base Sepolia key that anchors events
`BASE_SEPOLIA_RPC`	no	Default `https://sepolia.base.org`
`DEMO_TOKEN`	no	Real chain only for requests carrying this token. Unset means never real
`ZISHI_INTENT_LOG`	no	IntentLog address for real mode. Default: the Base Sepolia deployment in `contracts/deployments/base-sepolia.json`

Without any variable the app runs fully: scripted explanations, simulated chain, labelled as such. See `web/.env.example`.

14. Testing

Command	Where	Checks
`pnpm lint`	`web`	ESLint, Next.js config
`npx tsc --noEmit`	`web`	Types, including both dictionaries
`pnpm test`	`web`	The engine self check (section 11), including every learned metric assertion
`pnpm bench`	`web`	The before and after table in section 6.4, recomputed from the seeds
`pnpm charts`	`web`	The nine charts in `docs/charts`, redrawn from the seeds
`pnpm report`	`web`	`docs/EXECUTION_EVIDENCE.md` from the latest captured real run and the chain, `docs/RED_TEAM.md` from the attack set, `docs/EVALUATION.md` and `docs/results/` from the learned components
`forge test`	`contracts`	68 tests: `IntentLog.t.sol` (5), `Bond.t.sol` (28, three fuzzed), `JobBoard.t.sol` (11), `Attacks.t.sol` (15, the contract layer attack cases), `Lease.t.sol` (9). Needs `forge-std` in `contracts/lib`
`pnpm evidence && pnpm verify`	`web`	A full run and its balance sheet

CI (`.github/workflows/ci.yml`) runs lint, type check, self check, build and the Foundry tests on every push and pull request.

15. Security boundaries

Boundary	Rule
Messages	Data, never instructions. Framed, escaped, gated. Payees, amounts, caps and locations never come from text
Model	No key, no chain, no signing. Writes sentences, reads fields. Numbers are checked against facts
Intents	Typed, signed, nonce, replay checked. Every intent goes to the log and to every party before the policy engine decides
Policy engine	Code, not prompts. Rules are data. The model never sees it
Operator key	Only `set_cap` and `top_up`, only from the button, tier L3
Real chain	Only with the demo token and a funded key
Public routes	Best effort rate limit per instance, body limits, no storage; the demo token guards the funded key
Secrets	`.env*` ignored by git; `.env.example` has no values

What is simulated

Layer	Status
Contracts	Tested with Foundry, deployed and source verified on Base Sepolia. In real mode the engine calls IntentLog; Bond, JobBoard and Lease run in the mirrors
Chain	Simulated by default. Real mode commits every intent hash to the deployed IntentLog and anchors money events as calldata on Base Sepolia

Layer	Status
Keys	Generated per boot for seven actors. Production: secure element, hardware button, HSMs
Telemetry and confirmations	Signed by demo keys. Production: the drone's secure element and the farmer's app
Training data	Handwritten corpus and synthetic flights, labelled as such everywhere they appear
Money	Yuan as demo units. Production: bank money on a consortium chain (FISCO BCOS), no public token
Insurance and physical damage	Not in the build. Named as the one human decision path

Risks, what is built against each and what production would do are in `SELF_ASSESSMENT.md`.

16. By the numbers

Measured on 2026-10-20 from the self check, the bench, the charts script and Base Sepolia receipts.

Measure	Value	Source
Fleet scenario, simulated, fast	293 events in 1.3 s: 15 signed intents, 15 policy decisions over 113 rule lines, 25 transactions, 7 gate verdicts, 3 flight logs scored, 1 proof	`scenarios.ts`
Fleet scenario, real chain	15 `IntentLog.commit` calls, about 30 anchors, 0.000013 ETH	Base Sepolia receipts
Contract deployment	4 contracts, 4,491,890 gas, block 48,335,911	`contracts/deployments/base-sepolia.json`
Foundry	68 tests, 3 fuzzed at 256 runs	`forge test`
Attack set	52 cases, 7 benign, naive baseline falls to 51 of 52	`attacks.ts`
Classifier	349 messages, none of them the 52 attack texts, 87 held out, 2^14 hashed buckets	`ml/gate.ts`, `ml/corpus.ts`
Oracle	2,000 flights, 312 tampered, 750 fit, 634 honest calibration, 500 held out	`ml/plausibility.ts`
MDP	46,800 states, 26 weeks, 200 seasons	`economics/mdp.ts`
Stress test	2,000 seasons, 7 skip rates on the chart	`economics/stress.ts`
Drift	500 benign then 100 attack scores, 600 quiet	`drift/eval.ts`
Code	10,294 lines: engine 4,921, interface 3,179, Solidity 593, API routes 300, dictionaries 1,300	`wc -l`